

Probability & Random Processes

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2.6

Conditional Probability, Independence, Product Rule

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Recap (Probability)

Definition 2.9:

The **probability** of an event A is the sum of the weights of all sample points in A . Therefore,

$$0 \leq P(A) \leq 1, \quad P(\phi) = 0, \quad \text{and} \quad P(S) = 1.$$

Furthermore, if $A_1, A_2, A_3, \dots$ is a sequence of mutually exclusive events, then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots.$$



Recap (Probability)

- Many ways to assign probability of occurrence of the sample points of a sample space.
 - The indifference approach (No subjectivity, non information or intuition) (assume equally likely)
 - The use of intuition, personal beliefs, and other indirect information in arriving at probabilities is referred to as the subjective definition of probability.
 - the relative frequency interpretation of probability

Rule 2.3:

If an experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then the probability of event A is

$$P(A) = \frac{n}{N}.$$



Recap (Additive Rule)

Theorem 2.7

If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Corollary 2.1: If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

Corollary 2.2: If $A_1, A_2, \dots, A_n$ are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n).$$

Corollary 2.3: If $A_1, A_2, \dots, A_n$ is a partition of sample space S , then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) = P(S) = 1.$$

Theorem 2.8: For three events A, B , and C ,

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C). \end{aligned}$$



Recap (Additive rule)

Theorem 2.9

If A and A' are complementary events, then

$$P(A) + P(A') = 1.$$



Conditional Probability

The probability of an event B occurring
when it is known that some event A has occurred
is called a **conditional probability**
and is denoted by $P(B|A)$.

The symbol $P(B|A)$ is usually read
“the probability that B occurs given that A occurs”
or simply “the probability of B , given A .”



Conditional Probability

Consider the event B of getting a perfect square when a die is tossed.

The die is constructed so that the even numbers are twice as likely to occur as the odd numbers.

Based on the sample space $S = \{1, 2, 3, 4, 5, 6\}$, with probabilities of $1/9$ and $2/9$ assigned, respectively, to the odd and even numbers, the probability of B occurring is $1/3$.

Now suppose that it is known that the toss of the die resulted in a number greater than 3.



We are now dealing with a reduced sample space $A = \{4, 5, 6\}$, which is a subset of S .

To find the probability that B occurs, relative to the space A , we must first assign new probabilities to the elements of A proportional to their original probabilities such that their sum is 1.

Assigning a probability of w to the odd number in A and a probability of $2w$ to the two even numbers, we have $5w = 1$, or $w = 1/5$.

Relative to the space A , we find that B contains the single element 4.

Denoting this event by the symbol $B|A$, we write $B|A = \{4\}$, and hence

$$P(B|A) = \frac{2}{5}.$$

This example illustrates that events may have different probabilities when considered relative to different sample spaces.



We can also write

$$\begin{aligned}
 P(B|A) &= \frac{2}{5} \\
 &= \frac{2/9}{5/9} \\
 &= \frac{P(A \cap B)}{P(A)},
 \end{aligned}$$

where $P(A \cap B)$ and $P(A)$ are found from the original sample space S .

In other words, a conditional probability relative to a subspace A of S may be calculated directly from the probabilities assigned to the elements of the original sample space S .

Definition 2.10:

The conditional probability of B , given A , denoted by $P(B|A)$, is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0.$$



Suppose that our sample space S is the population of adults in a small town who have completed the requirements for a college degree. We shall categorize them according to gender and employment status.

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

One of these individuals is to be selected at random for a tour throughout the country

M : a man is chosen,

E : the one chosen is employed.

Using the reduced sample space E , we find that

$$P(M|E) = \frac{460}{600} = \frac{23}{30}.$$



Let $n(A)$ denote the number of elements in any set A . Using this notation, since each adult has an equal chance of being selected, we can write

$$P(M|E) = \frac{n(E \cap M)}{n(E)} = \frac{n(E \cap M)/n(S)}{n(E)/n(S)} = \frac{P(E \cap M)}{P(E)},$$

where $P(E \cap M)$ and $P(E)$ are found from the original sample space S .

we can write

$$P(M|E) = \frac{n(E \cap M)}{n(E)} = \frac{n(E \cap M)/n(S)}{n(E)/n(S)} = \frac{P(E \cap M)}{P(E)},$$

where $P(E \cap M)$ and $P(E)$ are found from the original sample space S .

To verify

$$P(E) = \frac{600}{900} = \frac{2}{3} \quad \text{and} \quad P(E \cap M) = \frac{460}{900} = \frac{23}{45}$$

$$P(M|E) = \frac{23/45}{2/3} = \frac{23}{30},$$

as before.



1. There are 500 students in a certain school. 150 students are enrolled in an Algebra course and 80 students are enrolled in a Chemistry course. There are 30 students who are taking both Algebra and Chemistry. If a student is chosen at random, (a) What is the probability that the student is taking Algebra? (b) What is the probability that the student is taking Chemistry given that the student is also taking Algebra? (c) What is the probability that the student is taking Algebra given that the student is also taking Chemistry?



Example 2.34:

The probability that a regularly scheduled flight departs on time is $P(D) = 0.83$;
the probability that it arrives on time is $P(A) = 0.82$;

and the probability that it departs and arrives on time is $P(D \cap A) = 0.78$.

Find the probability that a plane

- (a) arrives on time, given that it departed on time, and
- (b) departed on time, given that it has arrived on time.

Solution:

Using Definition 2.10, we have the following.

- (a) The probability that a plane arrives on time, given that it departed on time, is

$$P(A|D) = \frac{P(D \cap A)}{P(D)} = \frac{0.78}{0.83} = 0.94.$$

- (b) The probability that a plane departed on time, given that it has arrived on time, is

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.78}{0.82} = 0.95.$$



Example 2.35:

Consider an industrial process in the textile industry in which strips of a particular type of cloth are being produced.

These strips can be defective in two ways, length and nature of texture.

For the case of the latter, the process of identification is very complicated.

It is known from historical information on the process that 10% of strips fail the length test, 5% fail the texture test, and only 0.8% fail both tests.

If a strip is selected randomly from the process and a quick measurement identifies it as failing the length test, what is the probability that it is texture defective?

Solution: Consider the events

L : length defective, T : texture defective.

Given that the strip is length defective, the probability that this strip is texture defective is given by

$$P(T|L) = \frac{P(T \cap L)}{P(L)} = \frac{0.008}{0.1} = 0.08.$$

Thus, knowing the conditional probability provides considerably more information than merely knowing $P(T)$.



Independent Events

In the die-tossing experiment discussed on page 62, we note that $P(B|A) = 2/5$ whereas $P(B) = 1/3$.

That is, $P(B|A) \neq P(B)$, indicating that B depends on A .

Now consider an experiment in which 2 cards are drawn in succession from an ordinary deck, with replacement.

The events are defined as

A : the first card is an ace,

B : the second card is a spade.

Since the first card is replaced, our sample space for both the first and the second draw consists of 52 cards, containing 4 aces and 13 spades.

Hence,
$$P(B|A) = \frac{13}{52} = \frac{1}{4} \quad \text{and} \quad P(B) = \frac{13}{52} = \frac{1}{4}.$$

That is, $P(B|A) = P(B)$.

When this is true, the events A and B are said to be **independent**.



Although conditional probability allows for an alteration of the probability of an event in the light of additional material,

it also enables us to understand better the very important concept of **independence** or, in the present context, independent events.

In the airport illustration in Example 2.34, $P(A|D)$ differs from $P(A)$.

$$P(A|D) = 0.94. \quad P(A) = 0.82;$$

This suggests that the occurrence of D influenced A , and this is certainly expected in this illustration.

However, consider the situation where we have events A and B and

$$P(A|B) = P(A).$$

In other words, the occurrence of B had no impact on the odds of occurrence of A . Here the occurrence of A is independent of the occurrence of B .

The importance of the concept of independence cannot be overemphasized.

It plays a vital role in material in virtually all chapters in this book and in all areas of applied statistics.



Definition 2.11:

Two events A and B are **independent** if and only if

$$P(B|A) = P(B) \quad \text{or} \quad P(A|B) = P(A),$$

assuming the existences of the conditional probabilities. Otherwise, A and B are **dependent**.

The condition $P(B|A) = P(B)$ implies that $P(A|B) = P(A)$, and conversely.

For the card-drawing experiments, where we showed that $P(B|A) = P(B) = 1/4$, we also can see that $P(A|B) = P(A) = 1/13$.



The Product Rule, or the Multiplicative Rule

Multiplying the formula in Definition 2.10 by $P(A)$, we obtain the following important **multiplicative rule** (or **product rule**), which enables us to calculate the probability that two events will both occur.

Theorem 2.10:

If in an experiment the events A and B can both occur, then

$$P(A \cap B) = P(A)P(B|A), \text{ provided } P(A) > 0.$$

Thus, the probability that both A and B occur is equal to the probability that A occurs multiplied by the conditional probability that B occurs, given that A occurs.

Since the events $A \cap B$ and $B \cap A$ are equivalent, it follows from Theorem 2.10 that we can also write

$$P(A \cap B) = P(B \cap A) = P(B)P(A|B).$$

In other words, it does not matter which event is referred to as A and which event is referred to as B .



Example 2.36:

Suppose that we have a fuse box containing 20 fuses, of which 5 are defective. If 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?

Solution:

We shall let A be the event that the first fuse is defective and B the event that the second fuse is defective; then we interpret $A \cap B$ as the event that A occurs and then B occurs after A has occurred.

The probability of first removing a defective fuse is $1/4$; then the probability of removing a second defective fuse from the remaining 4 is $4/19$.

Hence,

$$P(A \cap B) = \left(\frac{1}{4}\right) \left(\frac{4}{19}\right) = \frac{1}{19}.$$



Example 2.37:

One bag contains 4 white balls and 3 black balls, and a second bag contains 3 white balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?

Solution:

Let B_1 , B_2 , and W_1 represent, respectively, the drawing of a black ball from bag 1, a black ball from bag 2, and a white ball from bag 1.

We are interested in the union of the mutually exclusive events $B_1 \cap B_2$ and $W_1 \cap B_2$.



The various possibilities and their probabilities are illustrated in Figure 2.8.

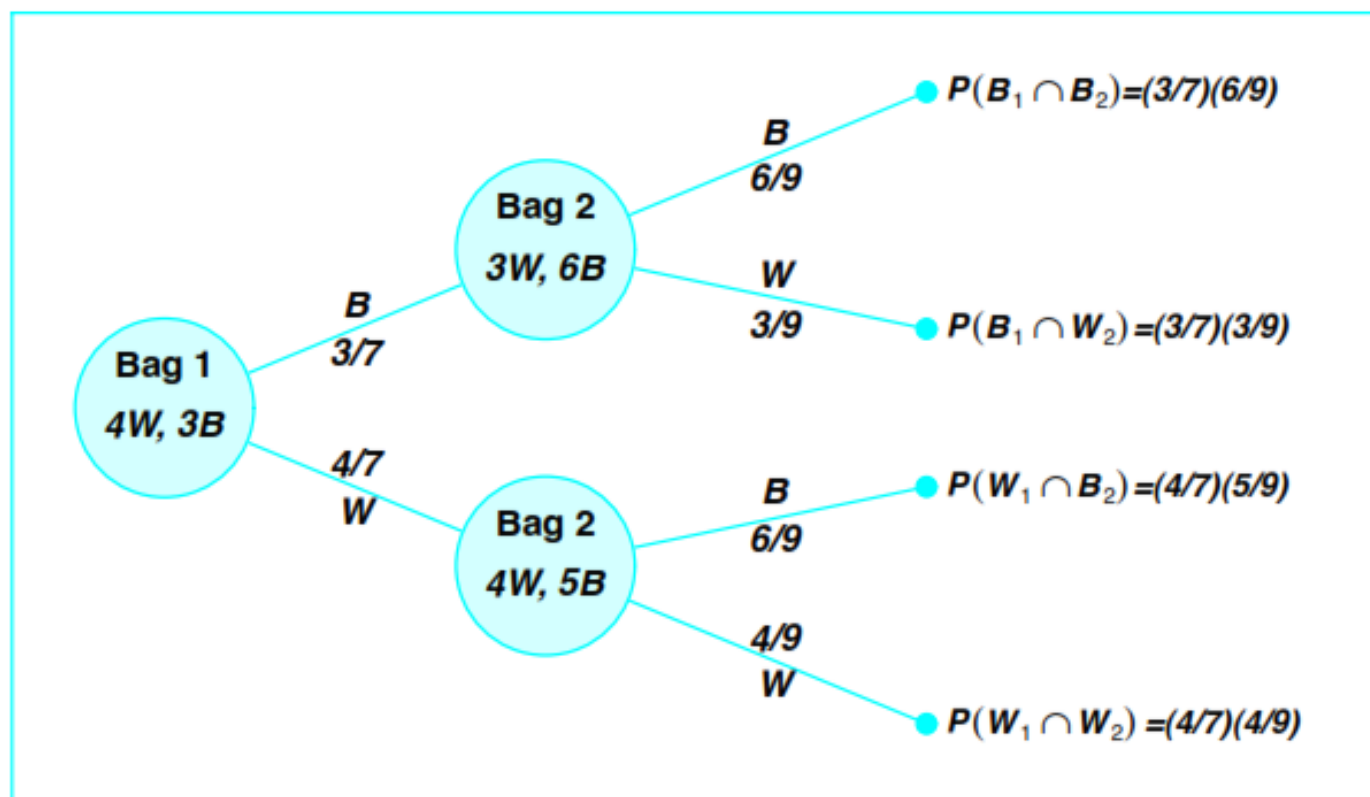


Figure 2.8: Tree diagram for Example 2.37.

$$\begin{aligned}
 \text{Now } P[(B_1 \cap B_2) \text{ or } (W_1 \cap B_2)] &= P(B_1 \cap B_2) + P(W_1 \cap B_2) \\
 &= P(B_1)P(B_2|B_1) + P(W_1)P(B_2|W_1) \\
 &= \left(\frac{3}{7}\right) \left(\frac{6}{9}\right) + \left(\frac{4}{7}\right) \left(\frac{5}{9}\right) = \frac{38}{63}.
 \end{aligned}$$



If, in Example 2.36, the first fuse is replaced and the fuses thoroughly rearranged before the second is removed, then the probability of a defective fuse on the second selection is still $1/4$; that is, $P(B|A) = P(B)$ and the events A and B are independent.

When this is true, we can substitute $P(B)$ for $P(B|A)$ in Theorem 2.10 to obtain the following special multiplicative rule.

Theorem 2.11:

Two events A and B are independent if and only if

$$P(A \cap B) = P(A)P(B).$$

Therefore, to obtain the probability that two independent events will both occur, we simply find the product of their individual probabilities.



Example 2.38:

A small town has one fire engine and one ambulance available for emergencies. The probability that the fire engine is available when needed is 0.98, and the probability that the ambulance is available when called is 0.92. In the event of an injury resulting from a burning building, find the probability that both the ambulance and the fire engine will be available, assuming they operate independently.

Solution:

Let A and B represent the respective events that the fire engine and the ambulance are available.

Then $P(A \cap B) = P(A)P(B) = (0.98)(0.92) = 0.9016$.



Example 2.39:

An electrical system consists of four components as illustrated in Figure 2.9.

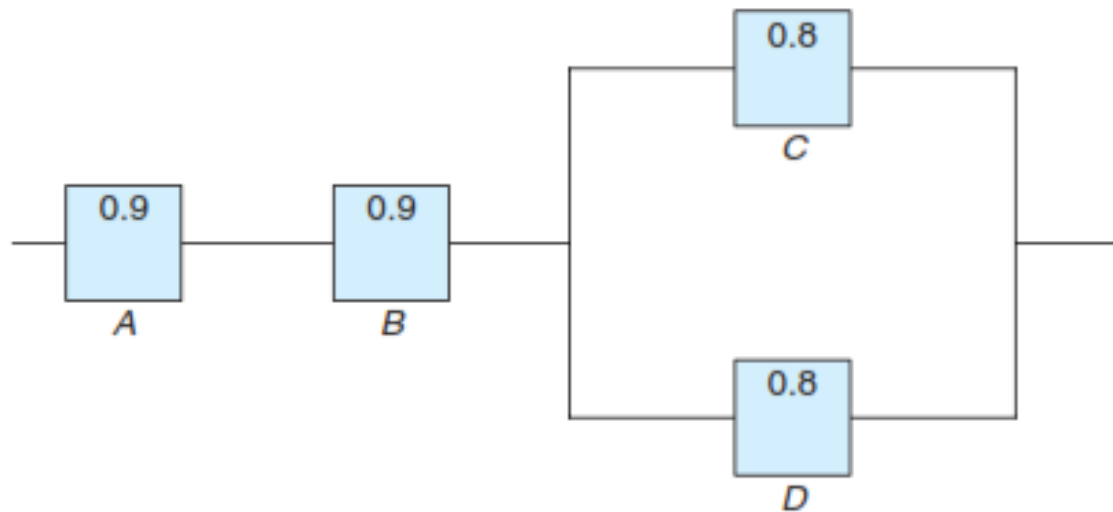


Figure 2.9: An electrical system for Example 2.39.

The system works if components A and B work and either of the components C or D works. The reliability (probability of working) of each component is also shown in Figure 2.9.

Find the probability that (a) the entire system works and (b) the component C does not work, given that the entire system works.

Assume that the four components work independently.

Solution:

In this configuration of the system, A , B , and the subsystem C and D constitute a serial circuit system, whereas the subsystem C and D itself is a parallel circuit system.

- (a) Clearly the probability that the entire system works can be calculated as follows:

$$\begin{aligned}P[A \cap B \cap (C \cup D)] &= P(A)P(B)P(C \cup D) = P(A)P(B)[1 - P(C' \cap D')] \\&= P(A)P(B)[1 - P(C')P(D')] \\&= (0.9)(0.9)[1 - (1 - 0.8)(1 - 0.8)] = 0.7776.\end{aligned}$$

The equalities above hold because of the independence among the four components.

- (b) To calculate the conditional probability in this case, notice that

$$\begin{aligned}P &= \frac{P(\text{the system works but } C \text{ does not work})}{P(\text{the system works})} \\&= \frac{P(A \cap B \cap C' \cap D)}{P(\text{the system works})} = \frac{(0.9)(0.9)(1 - 0.8)(0.8)}{0.7776} = 0.1667.\end{aligned}$$



The multiplicative rule can be extended to more than two-event situations.

Theorem 2.12:

If, in an experiment, the events $A_1, A_2, \dots, A_k$ can occur, then

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_k) \\ = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_k|A_1 \cap A_2 \cap \dots \cap A_{k-1}). \end{aligned}$$

If the events $A_1, A_2, \dots, A_k$ are independent, then

$$P(A_1 \cap A_2 \cap \dots \cap A_k) = P(A_1)P(A_2) \dots P(A_k).$$



Example 2.40:

Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards.

Find the probability that the event $A_1 \cap A_2 \cap A_3$ occurs, where A_1 is the event that the first card is a red ace, A_2 is the event that the second card is a 10 or a jack, and A_3 is the event that the third card is greater than 3 but less than 7.

Solution:

First we define the events

A_1 : the first card is a red ace,

A_2 : the second card is a 10 or a jack,

A_3 : the third card is greater than 3 but less than 7.

$$\text{Now } P(A_1) = \frac{2}{52}, \quad P(A_2|A_1) = \frac{8}{51}, \quad P(A_3|A_1 \cap A_2) = \frac{12}{50},$$

and hence, by Theorem 2.12,

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \\ &= \left(\frac{2}{52}\right) \left(\frac{8}{51}\right) \left(\frac{12}{50}\right) = \frac{8}{5525}. \end{aligned}$$



The property of independence stated in Theorem 2.11 can be extended to deal with more than two events.

Consider, for example, the case of three events A , B , and C .

It is not sufficient to only have that $P(A \cap B \cap C) = P(A)P(B)P(C)$ as a definition of independence among the three.

Suppose $A = B$ and $C = \phi$, the null set.

Although $A \cap B \cap C = \phi$, which results in $P(A \cap B \cap C) = 0 = P(A)P(B)P(C)$, events A and B are not independent.

Hence, we have the following definition.

Definition 2.12:

A collection of events $\mathcal{A} = \{A_1, \dots, A_n\}$ are mutually independent if for any subset of $\mathcal{A}$, $A_{i_1}, \dots, A_{i_k}$, for $k \leq n$, we have

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \dots P(A_{i_k}).$$

